ASSIGNMENT NO 9 SY

Title

Automation of Daily Report Generation from CSV Files Using Python

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Aim

To automate the process of loading CSV data, cleaning the dataset, generating summary reports, exporting results in CSV and PDF formats, and scheduling the task using Python (Cron / Task Scheduler).

Objective

1. To read and process CSV files using Python.
2. To perform data cleaning operations such as handling missing values and duplicates.
3. To generate summary statistics and reports automatically.
4. To export processed data into CSV and PDF formats.
5. To schedule automated execution using system schedulers like Cron (Linux) or Task Scheduler (Windows).

Prerequisites

- Basic knowledge of Python programming
- Understanding of CSV file structure
- Python installed (Version 3.x)
- Required Python Libraries:
    - pandas
    - numpy
    - matplotlib
    - reportlab (for PDF generation)

- Basic knowledge of:
    - Cron jobs (Linux)
    - Windows Task Scheduler

Theory

1. CSV File Handling

CSV (Comma Separated Values) files store tabular data in plain text format. In Python, the **pandas** library is widely used to read, manipulate, and analyze CSV data efficiently.

Example:

```
import pandas as pd
data = pd.read_csv("data.csv")
```

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2. Data Cleaning

Data cleaning is the process of preparing raw data for analysis. Common steps include:

- Removing null/missing values
- Filling missing values with mean/median
- Removing duplicate rows
- Correcting data types

Example:

```
data.dropna(inplace=True)
data.drop_duplicates(inplace=True)
```

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3. Generating Summary Reports

Summary reports provide statistical insights such as:

- Mean
- Median

- Standard Deviation
- Total and Count

Using:

summary = data.describe()

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4. Exporting Data

a) Save as CSV

summary.to_csv("summary_report.csv")

b) Save as PDF

Using **reportlab** library, formatted PDF reports can be generated programmatically.

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5. Task Scheduling

Cron Job (Linux/Mac)

Cron is a time-based job scheduler.

Command:

crontab -e

Example:

0 9 * * * /usr/bin/python3 /home/user/report_script.py

(Runs daily at 9:00 AM)

Windows Task Scheduler

1. Open Task Scheduler
2. Click "Create Basic Task"
3. Set Trigger (Daily)
4. Choose Action → Start Program
5. Select python.exe

6. Add script path in arguments

Procedure

1. Install required libraries using:

pip install pandas numpy matplotlib reportlab

2. Create a Python script for:
    - Reading CSV file
    - Cleaning data
    - Generating summary
    - Saving CSV and PDF
3. Test the script manually.
4. Configure Cron (Linux) or Task Scheduler (Windows).
5. Verify automated daily report generation.

Applications

- Business sales reporting
- Attendance tracking systems
- Financial report automation
- Inventory management
- Data analytics workflows

Advantages

- Saves time and manual effort
- Reduces human error
- Ensures consistent reporting
- Enables timely decision-making

Conclusion

In this experiment, we successfully automated the generation of daily reports from CSV files using Python. The system performs data loading, cleaning, summary generation, and exports results into CSV and PDF formats. By integrating task scheduling using Cron or Windows Task Scheduler, the reporting process becomes fully automated, efficient, and reliable. This approach enhances productivity and supports data-driven decision-making in organizations.